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OPERATIONS RESEARCH 415 PART IV: QUEUING THEORY

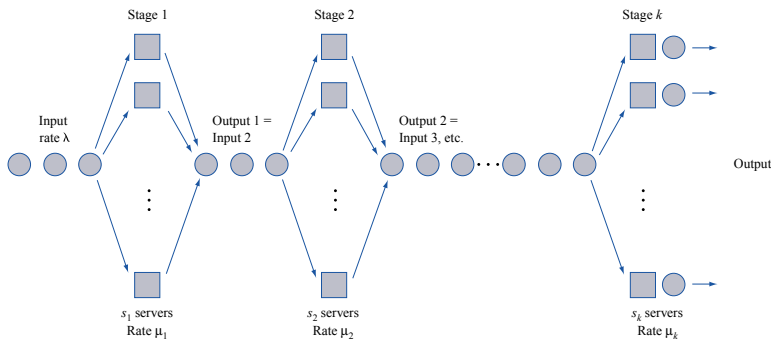
LECTURE 32: QUEUES IN SERIES AND OPEN QUEUING NETWORKS

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Exponential Queues in Series (pp. 1104–1105)

In many situations (such as the production of an item on an assembly line), the customer's service is not complete until he has been served by more than one server.



Exponential Queues in Series (pp. 1104–1105)

Upon entering the system, the arrival undergoes stage 1 service (after waiting in line if all stage 1 servers are busy on arrival).

After completing stage 1 service, the customer waits for and undergoes stage 2 service.

This process continues until the customer completes stage k service and the system is called a **k -stage series queuing system**.

Theorem 4: Jackson's Theorem (p. 1104)

If (1) inter-arrival times for a series queuing system are exponentially distributed with rate λ , (2) service times for the stage i server are exponentially distributed, and (3) each stage has an infinite-capacity waiting room, then inter-arrival times for arrivals to each stage of the queuing system are exponentially distributed with rate λ .

Exponential Queues in Series (pp. 1104–1105)

For Jackson's Theorem to hold, each stage must have sufficient capacity to service a stream of arrivals that arrives at rate λ ; otherwise, the queue will “blow up” at the stage with insufficient capacity.

From our discussion of the $M/M/s/GD/\infty/\infty$ queuing system, each stage will have sufficient capacity to handle an arrival stream of rate λ if and only if $\lambda < s_j\mu_j$ for all $j = 1, 2, \dots, k$.

If $\lambda < s_j\mu_j$, Jackson's Theorem implies that stage j of the system may be analyzed as an $M/M/s_j/GD/\infty/\infty$ system with exponential inter-arrival times having rate λ and exponential service times with a mean service time of $1/\mu_j$.

Auto Assembly (Example 13, p. 1105)

The last two things that are done to a car before its manufacture is complete are installing the engine and putting on the tires.

An average of 54 cars per hour arrive requiring these two tasks.

One worker is available to install the engine and can service an average of 60 cars per hour.

After the engine is installed, the car goes to the tire station and waits for its tires to be attached.

Three workers serve at the tire station. Each works on one car at a time and can put tires on a car in an average of 3 minutes.

Both interarrival times and service times are exponentially distributed.

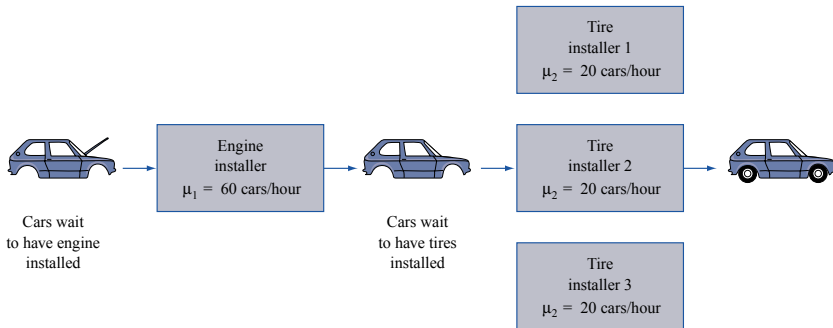
- 1 Determine the mean queue length at each work station.
- 2 Determine the total expected time that a car spends waiting for service.

Auto Assembly (Example 13, p. 1105)

$\lambda = 54$ cars/hour

Stage 1

Stage 2



An **open queuing network** is a generalisation of queues in series.

Assume that station j consists of s_j servers with exponential service time, each operating at rate μ_j .

Customers are assumed to arrive at station j from outside the queuing system at a rate r_j . These inter-arrival times are exponentially distributed.

Once completing service at station i , a customer joins the queue at station j with probability p_{ij} and completes service with probability $1 - \sum_{j=1}^k p_{ij}$.

Let λ_j be the rate at which customers arrive at station j (this includes arrivals at station j from outside the system *and* from other stations).

Then $\lambda_1, \lambda_2, \dots, \lambda_k$ can be found by solving the system of linear equations

$$\lambda_j = r_j + \sum_{i=1}^k p_{ij} \lambda_i, \quad j = 1, 2, \dots, k.$$

Then it can be shown that the probability distribution of the number of customers present at station j and the expected number of customers present at station j can be found by treating station j as an $M/M/s_j/GD/\infty/\infty$ system with arrival rate λ_j and service rate μ_j .

If, for some j , $\lambda_j \geq s_j \mu_j$, then no steady-state distribution of customers exists.

Remarkably, the numbers of customers present at each station are independent random variables. (That is, knowledge of the number of people at all stations other than station j tells us nothing about the distribution of the number of people at station j !) **This result does not hold, however, if either inter-arrival or service times are not exponential.**

To find L , the expected number of customers in the queuing system, simply add up the expected number of customers present at each station.

To find W , the average time a customer spends in the system, simply apply Little's Formula $L = \lambda W$ to the entire system, where $\lambda = r_1 + r_2 + \cdots + r_k$.

Open Queuing Network (Example 14, p. 1107)

Consider two servers. An average of 8 customers per hour arrive from outside at server 1, and an average of 17 customers per hour arrive from outside at server 2.

Inter-arrival times are exponentially distributed.

Server 1 can serve at an exponential rate of 20 customers per hour, and server 2 can serve at an exponential rate of 30 customers per hour.

After completing service at server 1, half of the customers leave the system, and half go to server 2. After completing service at server 2, 75% of the customers complete service, and 25% return to server 1.

- 1 What fraction of the time is server 1 idle?
- 2 Find the expected number of customers at each server.
- 3 Find the average time a customer spends in the system.
- 4 How would the answers to parts (1)–(3) change if server 2 could serve only an average of 20 customers per hour?